Bollinger Band Based Mean Reversion Trading Strategy Analysis & Backtesting Report

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Abstract

This study explores a mean reversion trading strategy built on Bollinger Bands, applied to SENSEX 30 and NIFTY 50 indices. Based on the idea that prices revert to their historical mean, the strategy generates buy signals when prices fall below the lower band and sell signals when they rise above the upper band. A 5% trailing stop loss was added to control drawdowns. Historical OHLCV data were analyzed using a custom Python backtesting framework designed to avoid lookahead bias and benchmarked against a buy-and-hold portfolio. Nine parameter combinations (n = 20, 50, 100; k = 1.5, 2.0, 2.5) were tested. The best configuration (n = 20, k = 1.5) yielded a 141% total return, Sharpe ratio of 1.21, and maximum drawdown of 9.36%, with a win/loss ratio of 2.73. Results showed that shorter lookback periods and narrower bands improved responsiveness and profitability, while the trailing stop loss reduced downside risk. Visual analyses through heatmaps and equity curves confirmed the robustness of these findings. Overall, the research demonstrates that a disciplined, statistically driven Bollinger Band mean reversion strategy can outperform passive investing by systematically converting volatility into opportunity. Future work could extend the model across asset classes and incorporate adaptive parameters or momentum filters for enhanced resilience.

Keywords— Bollinger Bands, Mean Reversion Strategy, Volatility-Based Trading, Quantitative Backtesting, Trailing Stop Loss, Parameter Optimization, Sharpe Ratio, Maximum Drawdown, Equity Curve Analysis, Statistical Signal Generation

1 Introduction

This report discusses in detail a mean reversion trading strategy that uses Bollinger Bands as the foundation of the analysis. The principle of mean reversion is one of the oldest and most persistent concepts in finance. It maintains that over time, the price and return of an asset will return to their historical averages. Periods of overvaluation and undervaluation come and go, and their price adjustments provide the systematic trader with opportunities if he or she can only operate with discipline and patience. Bollinger Bands, developed by John Bollinger, operationalize this principle by providing statistical boundaries around a moving average. The width of the bands dynamically changes as a function of volatility, which makes them very versatile in highlighting potential reversals. The following study explores how Bollinger Bands could be applied to develop a well-structured trading system-one that identifies opportunity, controls risk, and is viable through time. This analysis embeds theoretical insights, backtesting results, and interpretative discussion within an attempt to bridge academic rigor and practical application.

2 Objective

The core hypothesis of the present study is to back-test and validate the profitability and robustness of a mean reversion trading framework anchored in Bollinger Bands. The study aims to identify if it is possible to outperform a passive buy-and-hold investment by optimizing entry and exit timing with controlled exposure to risk. The investigation is focused on finding the optimal combination of two major parameters: the moving average lookback period, n, and the standard deviation multiplier, k, determining the sensitivity of the bands and, therefore, the frequency and precision of the trading signals. Moreover, the research discusses the importance of a trailing stop loss mechanism, introduced to protect the gains against deep drawdowns. By examining various configurations and different performance metrics, the study constructs a quantitative case for disciplined mean reversion as a sustainable means of portfolio management.
Features of the code:

1. Gets OHCLV data from yfinance.
2. Calculates 20-day SMA and Std. Deviation
3. Builds Upper and Lower Bands using configurable lookback (n) and width multiplier (k).
4. Generates trading signals based on Bollinger logic:
    - Buy when price crosses below the lower band.
    - Sell when price crosses above the upper band (or SMA).
5. Ensure trades execute the day after a signal (no lookahead bias)
6. Comprehensive visualizations
7. Backtesting engine tracking performance metrics
8. Comparison with Buy N Hold Strategy
9. Visualization of Optimization

3 Data and Tools

The data used in this research consists of daily historical price records of the SENSEX 30 and NIFTY 50 indices. Both have been chosen because of their high liquidity, representative nature,

and potential impact on the overall Indian financial market. Data included open, high, low, and close prices, sequentially organized to reflect temporal accuracy. Each dataset has been cleaned, checked for missing entries, and preprocessed to retain continuous data. All analytical computations have been performed in Python because the language offers efficiency and reproducibility. For handling data, doing statistical calculations, and generating signals, Pandas and NumPy were used. Matplotlib and Plotly were used for equity curves, returns, and parameter heatmaps in detail. The backtesting framework has been manually implemented rather than adapting any predefined libraries to keep full control over transaction flow, state management, and performance monitoring. This ensures that every variable, from position sizing to equity calculation, can be audited and adjusted with precision.

4 Strategy Design

The Bollinger Band strategy was constructed based on the mean, taken as a simple moving average over a window of n days. Above and below this mean, two more lines are drawn showing k standard deviations of price movement. The width expands with high volatility and reduces when markets stabilize. The system is based on three basic rules for trading. First, whenever the price falls below the lower Bollinger Band, that will create a buy signal, as at such times the price is oversold and is expected to revert to mean. Second, whenever the price rises above the upper Bollinger Band, that will trigger a sell signal, because the price is overbought and expected to experience a downturn reversion. Third, a trailing stop loss would be set at five percent below the highest price achieved since the buy signal was given to ensure that profits were locked in during an unexpected market reversal. To simplify tracking of exposure, the strategy assumed full capital deployment in each trade to test the intrinsic profitability of the system. The initial capital was one lakh rupees, and the simulation executed trades in chronological order to simulate how the real market would progress.

5 Backtesting Framework

The backbone of this research is the backtesting component, which has simulated a daily trading process using sequential data and has made sure that a trade was always based on previously known information to avoid lookahead bias. The five key variables that got updated in every step were current cash holdings, number of shares owned, daily portfolio equity, cumulative return, and position status (long or neutral). The framework also logged all trade events, including buy and sell dates, prices, and the reasons for exit. A benchmark portfolio was also used, applying a simple buy and hold strategy to the same data for direct comparison. Key performance indicators that were used include the total return, Sharpe ratio, maximum drawdown, win to loss ratio, and average winning upside. The total return gives an idea about the overall profitability, while the Sharpe ratio has accounted for volatility to examine the return consistency. Maximum drawdown expresses the largest percentage drop in equity, thus defining downside exposure. The win to loss ratio describes the efficiency of the trades, whereas the average upside wins show the size of the profitable movements the system had captured.

Math Involved
For each time t:

$$\text{SMA}_t = \frac{1}{n}\sum_{i=0}^{n-1} P_{t-i}$$

$$\text{SD}_t = \sqrt{\frac{1}{n-1}\sum_{i=0}^{n-1}(P_{t-i} - \text{SMA}_t)^2}$$

$$\text{UB}_t = \text{SMA}_t + k \cdot \text{SD}_t$$

$$\text{LB}_t = \text{SMA}_t - k \cdot \text{SD}_t$$

where:

- P_t = closing price at time t
- n= lookback window
- k = standard deviation multiplier (band width)
- SMA = Simple Movin Average
- SD = Standard Deviation
- UB = Upper Bollinger band
- LB = Lower Bollinger Band

6 Results

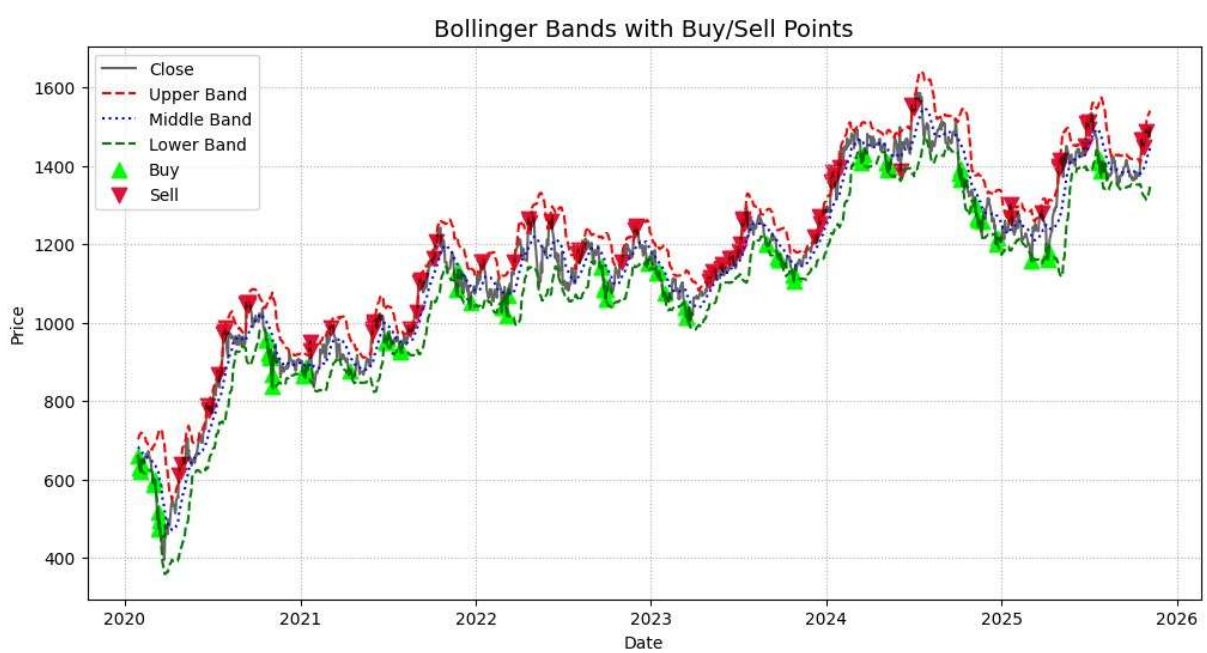
Testing was conducted across nine configurations formed by combining three values for n (20, 50, and 100) and three for k (1.5, 2.0, and 2.5). A similar pattern emerged, with shorter lookback periods and narrower bands generally performing better than longer/wider configurations. With one exception, the best-performing configuration returned 141 percent overall, had a Sharpe ratio of 1.21, and experienced only a 9.36 percent maximum drawdown. This configuration executed 41 trades and maintained a win to loss ratio of 2.73, meaning that for every losing trade there were nearly three profitable ones. Larger n or k values resulted in smoother, more gradual responses to market action with fewer trades and lower aggregate returns. The trailing stop loss mechanism reduced drawdowns across the board, confirming its status as a damping factor in protecting equity from abrupt reversals. The pattern across parameters reinforced the notion that responsiveness to recent market data increases profitability when combined with disciplined exit rules.

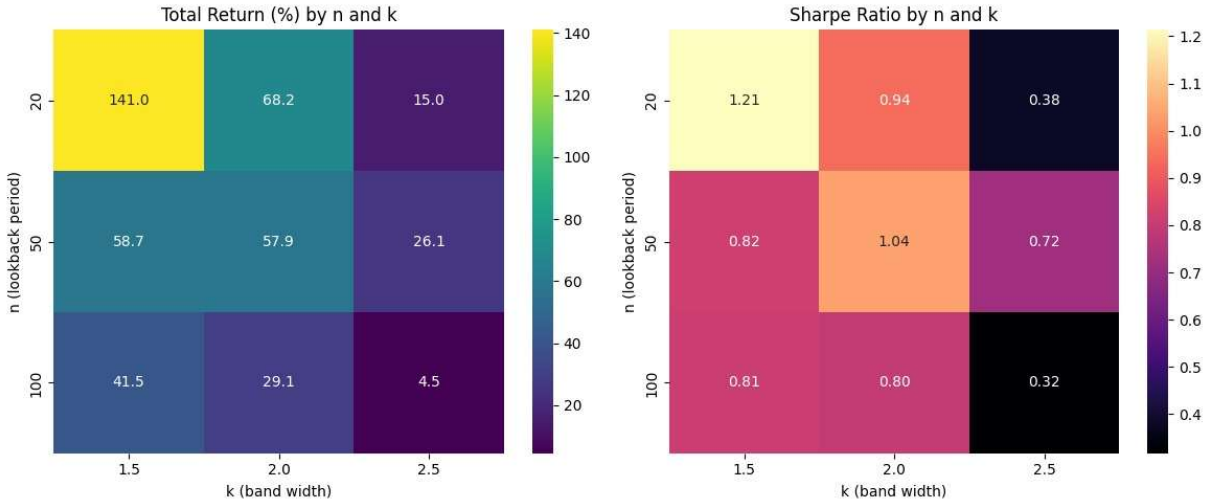
7 Interpretation

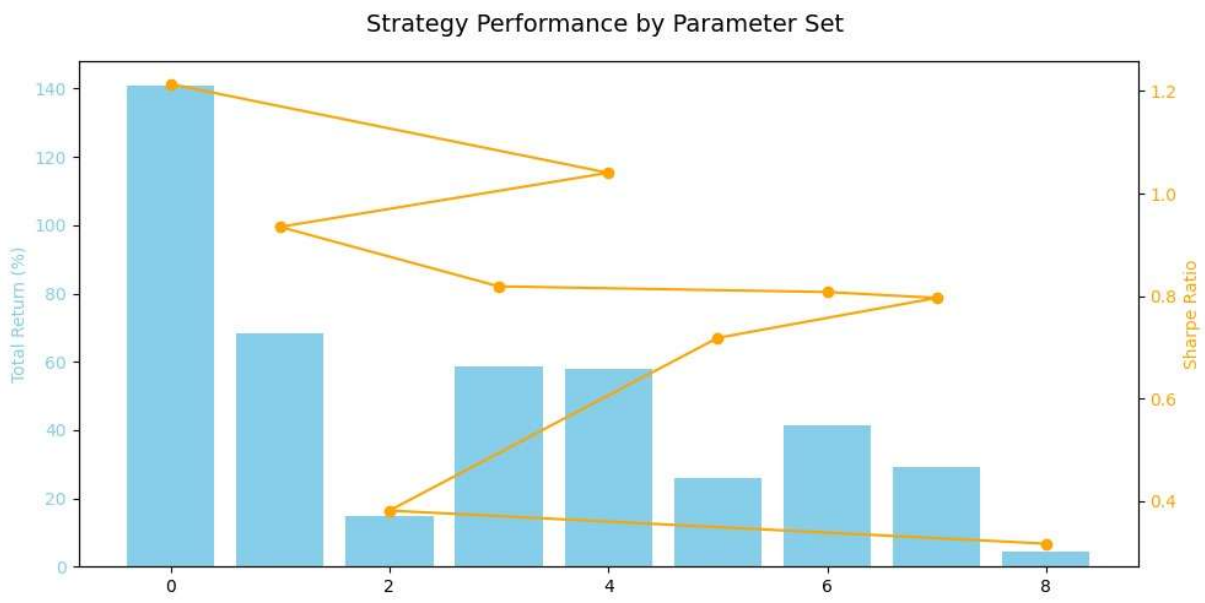
This analysis underlines the fact that mean reversion strategies prosper in markets with cyclical fluctuations around a stable mean; they do best in conditions where prices oscillate rather than strongly trend in one direction. Shorter lookback periods allowed the algorithm to respond quickly to deviations in an effective way, capturing small reversions before they evaporated. However, higher sensitivity also led to a higher trading frequency and, occasionally, false signals in periods of low volatility. The inclusion of a five percent trailing stop loss offset this drawback by ensuring that temporary missteps did not evolve into significant losses. Longer lookback periods and wider bands fitted delayed responses, missing short-term reversions and decreasing the return potential. These findings indicate that the ideal balance of speed and caution is achieved by calibrating parameters to the volatility profile of the underlying asset. This research confirms that volatility, while generally perceived as risk, can be tamed as a source of persistent opportunity when seen through the lens of statistical models, such as Bollinger Bands.

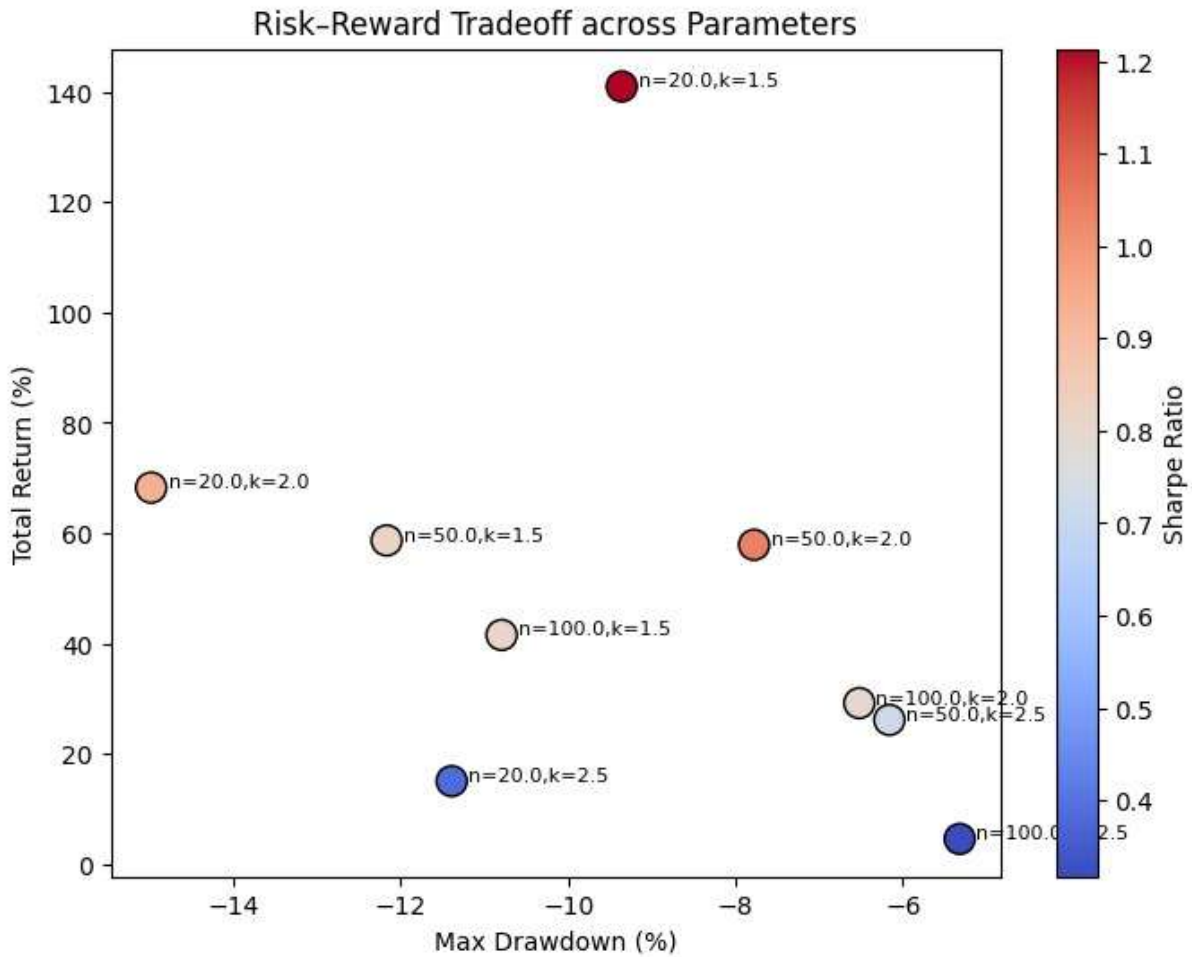
8 Visual Analysis

Visual analytics were critical in verifying the numerical results. Heatmaps representing total returns and Sharpe ratios across combinations of n and k showed a clear cluster of top performers centered around $n = 20$ and $1.5 \leq k \leq 2.0$. Scatterplots of total return vs. maximum drawdown showed the clustering of top performers in the top left quadrant, indicating high returns with controlled risk. The time series plots of equity curves showed how the strategy achieved steady growth with limited volatility, with smoother trajectories compared to the benchmark buy and hold. Supplementing this, additional histograms of daily returns supported that the strategy was able to cut down the downside tail while preserving upside potential. These visualizations not only strengthened statistical conclusions but also enhanced interpretability toward real-world application.




Total Return (%) by n and k
n (lookback period)
k (band width)
20
50
100
1.5
2.0
2.5
141.0
68.2
15.0
58.7
57.9
26.1
41.5
29.1
4.5
140
120
100
80
60
40
20
Sharpe Ratio by n and k
1.21
0.94
0.38
0.82
1.04
0.72
0.81
0.80
0.32
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4


Strategy Performance by Parameter Set
Total Return (%)
Sharpe Ratio
140
120
100
80
60
40
20
0
1.2
1.0
0.8
0.6
0.4
0
2
4
6
8



9 Conclusion

This research concludes that a structured mean reversion trading strategy using Bollinger Bands yields superior returns with manageable risk when it is implemented with discipline and properly optimized parameters. The adaptive band-based signals in conjunction with a trailing stop loss mechanism created a self-correcting system able to adapt to market conditions. The configuration with $n = 20$ and $k = 1.5$ showed the most robust balance between reactivity and reliability. By systematically transforming volatility into opportunity, the strategy exemplifies how quantitative logic can outperform intuition-based trading. The study reaffirms that success in financial markets depends not on prediction but on the consistent execution of statistically sound rules supported by rigorous testing. Recommendations This could be furthered by future research testing the scalability of the strategy across asset classes such as global equities, commodities, and currencies. Dynamic adjustment mechanisms, whereby k changes relative to the rolling volatility, would also serve to increase the strategy's responsiveness in rapidly changing markets. Additional filters, such as the Relative Strength Index or the Moving Average Convergence Divergence, could further reduce false signals and refine timing. The use of realistic assumptions about transaction costs, bid-ask spreads, and slippage will further increase the finesse associated with evaluating net performance. Lastly, future research could partition the data into bullish, bearish, and sideways regimes and thus provide another dimension to the robustness of a strategy by illustrating its behavior under different macroeconomic conditions. Summary This report shows that with good statistical underpinnings, a quantitative discipline can turn randomness into consistent opportunity. A Bollinger Band-based mean reversion strat-

egy transforms volatility from a threat to an asset. With clear rules, quantifiable controls of risk, and mechanical execution, the system minimizes mistakes born of emotion while maximizing capital efficiency. The heart of this research lies in the realization that success in finance does not come about through forecasting prices of the future but in forming systems that intelligently respond to change. The framework of the Bollinger Band, if used with patience, precision, and adaptability, stands as a functional example of how mathematics and discipline together can manufacture sustainable profitability in uncertain markets.